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Unprecedented Sampling Efficiency in Low Resistivity Pay with ORA Intelligent Wireline Formation Testing Platform for ADNOC Offshore

S. Menon, K. Alameri, and K. K. Nasr, ADNOC Offshore; M. Sarili, M. Yacoub, S. Al-Zubaidi, E. Kazakevich, S. Ashish, D. Ghosal, P. Raj, A. Jain, and V. Kumar, SLB Middle East

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Abstract

The formation evaluation along with reservoir fluid characterization are always the crucial objectives at any stage of field development cycle. Evaluating the presence of hydrocarbons in the low resistivity pay carbonate is more critical. Wireline formation testing helps to flow from each level to confirm formation fluid. The paper demonstrates the cutting-edge technology introduced to ADNOC Offshore for achieving all the mentioned objectives.

The limitations in petrophysical evaluation necessitate dynamic reservoir evaluation to accurately identify the formation fluid. In our case the target zone had low resistivity with multiple sub-layers which required to test each of them in deviated well with high sticking risk. The newly introduced radial focused inlet along with modernized pump, allowing to displace fluid with expanded range of flow rates depending on reservoir properties, made it possible to test multiple zones in the most efficient way. The latest generation in-situ fluid analyzers were exploited to obtain detailed fluid properties (hydrocarbon composition, density, viscosity etc.) in real-time during the acquisition.

The deployment of the latest intelligent wireline formation testing platform in the well allowed to perform testing at eight different depths and acquire critical reservoir information within 24 hours. The downhole fluid identification analysis indicated formation water at two depths and formation water with some fraction of oil at six stations. The embedded fluorescence detector showing high sensitivity to hydrocarbons proved its effectiveness in detecting the tiniest oil traces in the water dominated fluid flow. The enhanced hydrocarbon composition along with gas-oil ratio, density and viscosity was determined at two stations having high water cut (~80-90%) owing to sophisticated in-situ fluid analyzer. Six downhole fluid samples were collected during acquisition for subsequent laboratory analysis. One downhole Constant Composition Experiment (CCE) was performed to estimate the bubble point pressure at the station with 20% of oil fraction. Six Interval Pressure Transient Tests (IPTT) were carried out to evaluate the main reservoir properties like horizontal permeability and formation pressure. The successful implementation and acquired results proved the effectiveness of the selected technology and paved the way for future implementation in the more challenging downhole conditions.

The advantage of the latest intelligent wireline formation testing platform to achieve all objectives in multiple layer reservoirs within single wireline run lasting only 24 hours, which would be impossible with legacy formation testing tool require drill pipe conveyance deployment which would require ~5-6 days to evaluate. So, shorter formation testing time led to rig time reduction, lower electricity / fuel consumption and less CO₂ emission in the atmosphere.

Introduction

The field in discussion is an offshore field located in the Arabian Gulf at the northern edge of the territorial waters of the Emirate of Abu Dhabi. The field is situated 180 km away from Abu Dhabi. It makes up the southern sector of an oil accumulation.

The field was discovered in 1969 with an exploration well. Subsequent delineation was conducted in 1973 and 1974 with an initial oil production taking place in 1974.

The field has been producing oil from several reservoirs (Lower Cretaceous and Middle to Upper Jurassic).

Reservoir Description

The field is a large oval structure of NE-SW anticline, affected by NW-SE trending fault corridor (Fig. 1). The stratigraphic sequence is a thick calcareous platform succession from Permian to recent similar to other fields in the region.

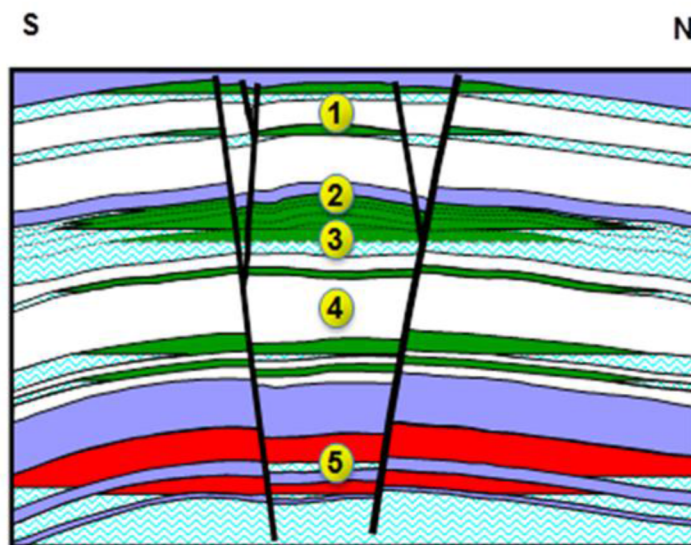


Figure 1—Field Cross-Section

Most common lithologies are limestone, dolomite, anhydrite and shale indicating shallow marine to supratidal depositional environment. Anhydrite is observed in the Upper Jurassic whereas minor clastics prevail in the Pre-Khuff series.

The top reservoir is a secondary reservoir of Lower Cretaceous age, the middle reservoirs are the main producing reservoirs of Upper Jurassic age and compromise two sub reservoirs the upper and the lower, while the bottom reservoir is a marginal reservoir from Middle Jurassic age. The underneath reservoir is a gas reservoir of Permian age.

The target for the ORA sampling was the Lower Cretaceous Unit. Within the Lower Cretaceous Unit, the primary target was Reservoir 5 and the secondary targets were Reservoirs II & IV (Fig. 2).

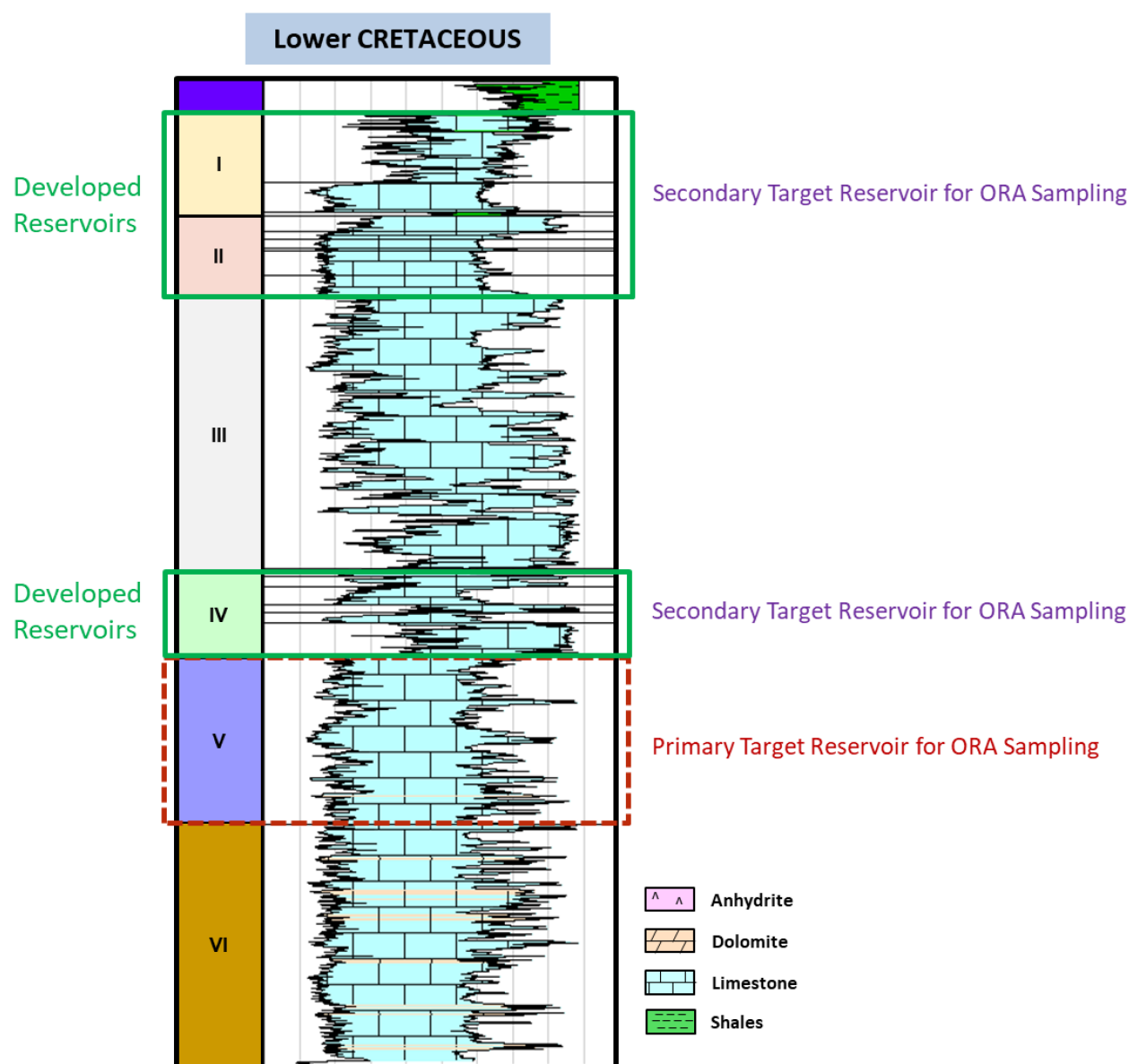


Figure 2—Lower Cretaceous Reservoir Electrical Log

The thickness of the primary target reservoir ranges between 350 – 400 ft. with very good porosity ranging between 24 – 30%. Based on the porosity, the reservoir is expected to vertical connectivity. Pressure points were also included in the program to confirm the same.

Well Selection & Design

A candidate well was selected for re-entry and sidetrack with a 6" horizontal drain in the Upper Jurassic Reservoir, as the existing drain was already at 99% BSW.

As this well was penetrating the lower cretaceous reservoirs in the crest, the opportunity was taken to conduct FID & Sampling to appraise Reservoir V which was never previously sampled or tested. The gas shows from crossing crestal wells was promising. This particular reservoir is being developed in a neighboring field. Along with this, it was also decided to perform FID & Sampling across Reservoirs II & IV to confirm no further potential in the crest.

The existing drain was P&A'ed, and the window was opened inside the 9-5/8" casing, and the 8½" Section was drilled all across the lower cretaceous unit and stopped at the thick anhydrite layer above the Upper Jurassic Unit.

Prior to deploying the ORA Tool, pressure points were recorded. The ORA Tool was then deployed on wireline to perform the FID & Sampling jobs across the target reservoirs. Based on the results of the job, there were 2 options ():

1. If Oil is sampled across the Lower Cretaceous Reservoir V, the 6" Drain would be drilled across this reservoir
2. If Water is sampled across the Lower Cretaceous Reservoir V, the well would be deepened to it's primary target in the Upper Jurassic Reservoir

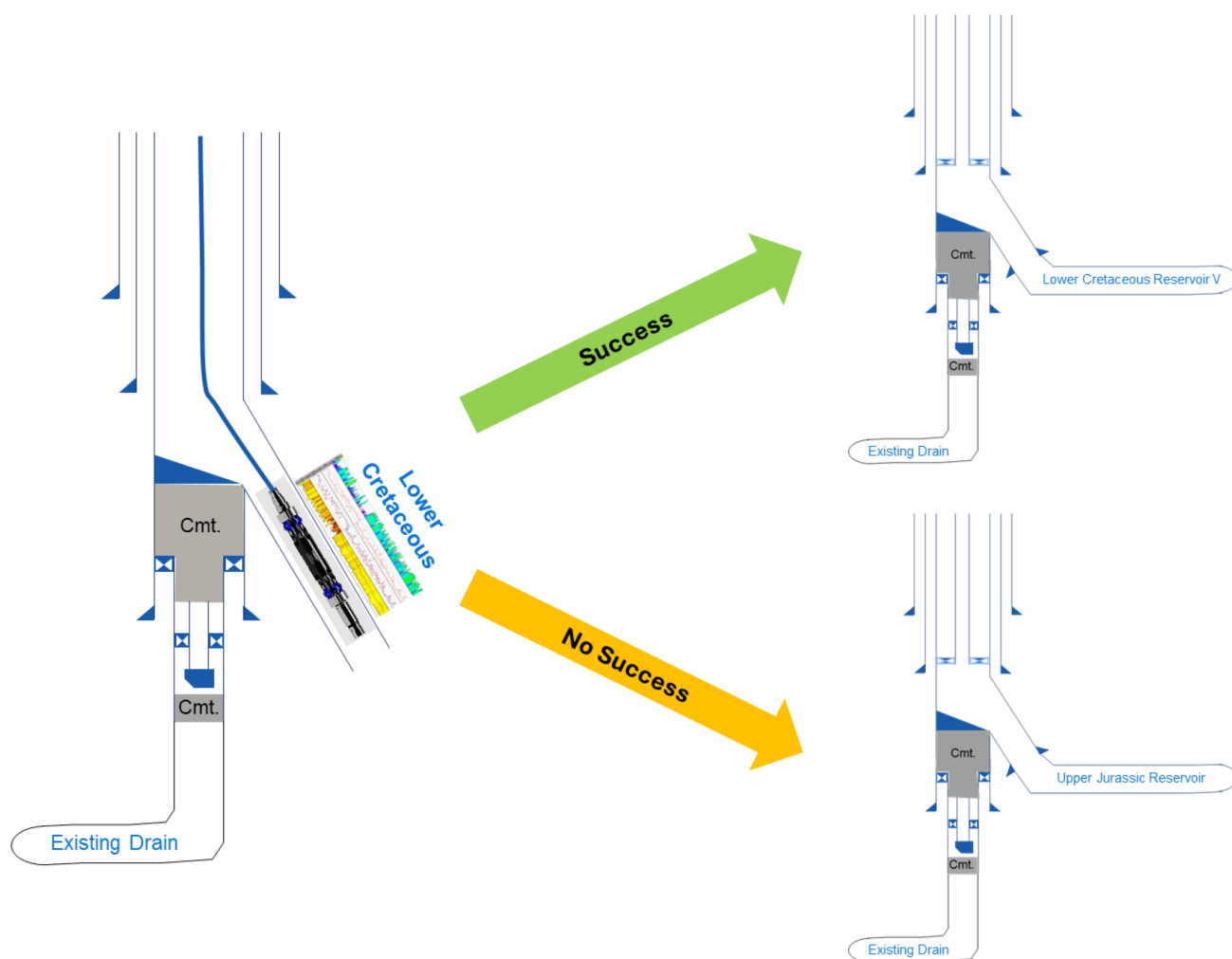


Figure 3—Well Design with 2 Options

To address the challenges of reservoir evaluation in complex carbonate formations, the ORA Intelligent Wireline Formation Testing Platform was introduced as a cutting-edge solution. This advanced technology is specifically designed to overcome the limitations of conventional formation testing tools. The ORA platform integrates a dual-flowline architecture, laboratory-grade metrology, and smart downhole automation, enabling precise fluid identification and dynamic reservoir evaluation in real time. Its ability to perform focused sampling and efficient clean-up, even in challenging environments, makes it an ideal choice for evaluating multilayered carbonate formations with low permeability. The following sections detail the

deployment of the ORA platform in a deviated well offshore Abu Dhabi, highlighting its role in achieving comprehensive reservoir characterization and fluid sampling with unprecedented efficiency.

The primary challenges in this operation included:

1. **Low Resistivity Pay Zones:** The low resistivity nature of the target formation made it difficult to distinguish between hydrocarbon-bearing and water-bearing intervals using conventional petrophysical methods.
2. **Low Permeability Formation:** The multilayered carbonate formation with permeability ranging from 1-5 mD required advanced tools capable of focused sampling and efficient clean-up and accurate fluid identification.
3. **Deviated Well with High Sticking Risk:** The well's deviation introduced a significant risk of sticking during stationary wireline operations, necessitating innovative solutions to ensure safe and efficient data acquisition. (See Fig. 4)

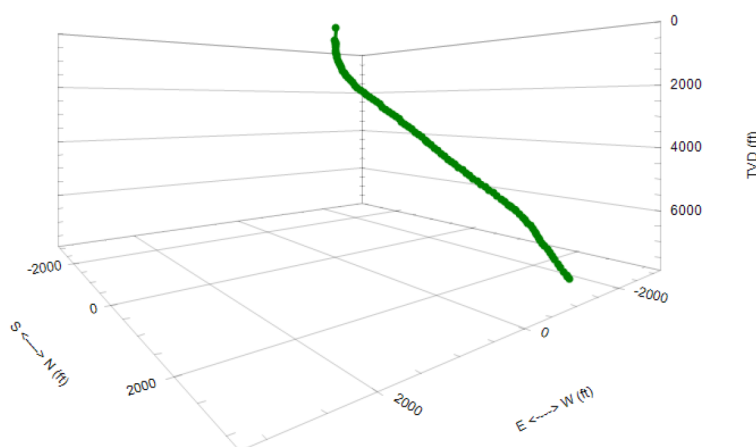


Figure 4—Well Profile

To address these challenges, the ORA Intelligent Wireline Formation Testing Platform was selected for its advanced capabilities, including:

- **Dual-Flowline Architecture:** Enabled efficient clean-up and focused sampling in low permeability formations. (See Fig. 5)
- **Laboratory-Accuracy Metrology:** Provided real-time fluid identification and detailed fluid properties, such as hydrocarbon composition, density, and viscosity. (See Fig. 6)
- **Smart Downhole Automation:** Optimized tool performance and minimized operational risks.
- **Intelligent Planning:** Clean up simulation to estimate station time, estimate volume of hydrocarbon to pump borehole and wellbore dynamic simulation to for safe operation.

Select Tools	Total Length (ft)	Flowline Sketch	Tool Sketch
LEH	92.54		
FNPC	89.05		
EDTCH	81.36		
FNKU_L1_L2_Closed	73.36		
FCRF_OFF7	72.16		
Tool Zero = Yes			
FNFM2_X_L1	57.46		
Select for Toolzero ?			
FNFM1_X_L2	42.65		
Select for Toolzero ?			
FISO	27.84		
Select for Toolzero ?			
FSCM_Drive_DN	16.50		
Select for Toolzero ?			
101442234 L1-8wde 101442236 Water-L2	Upper FRP Lower FRP		
FNKL_DN	1.09		

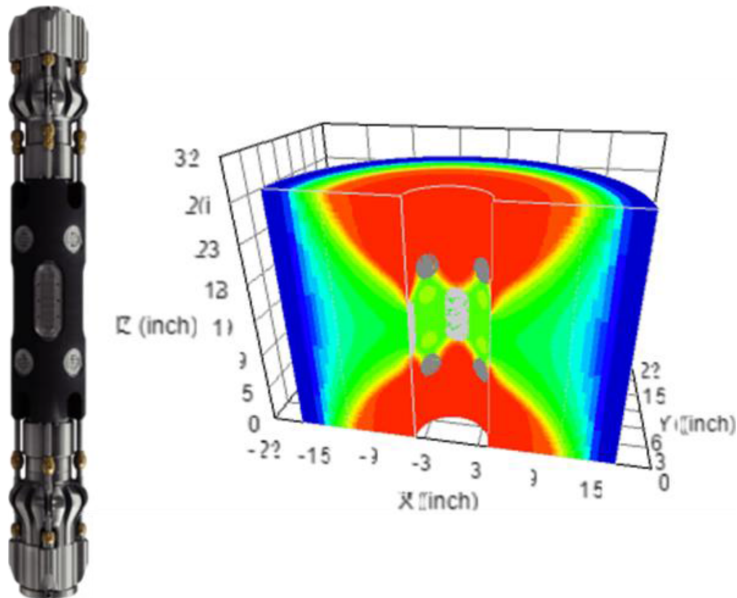


Figure 5—ORA Digital Hardware, optimized with Focused Radial Probe, Two Flow Manager, Fluid InSitu Scanner and Sample Manager

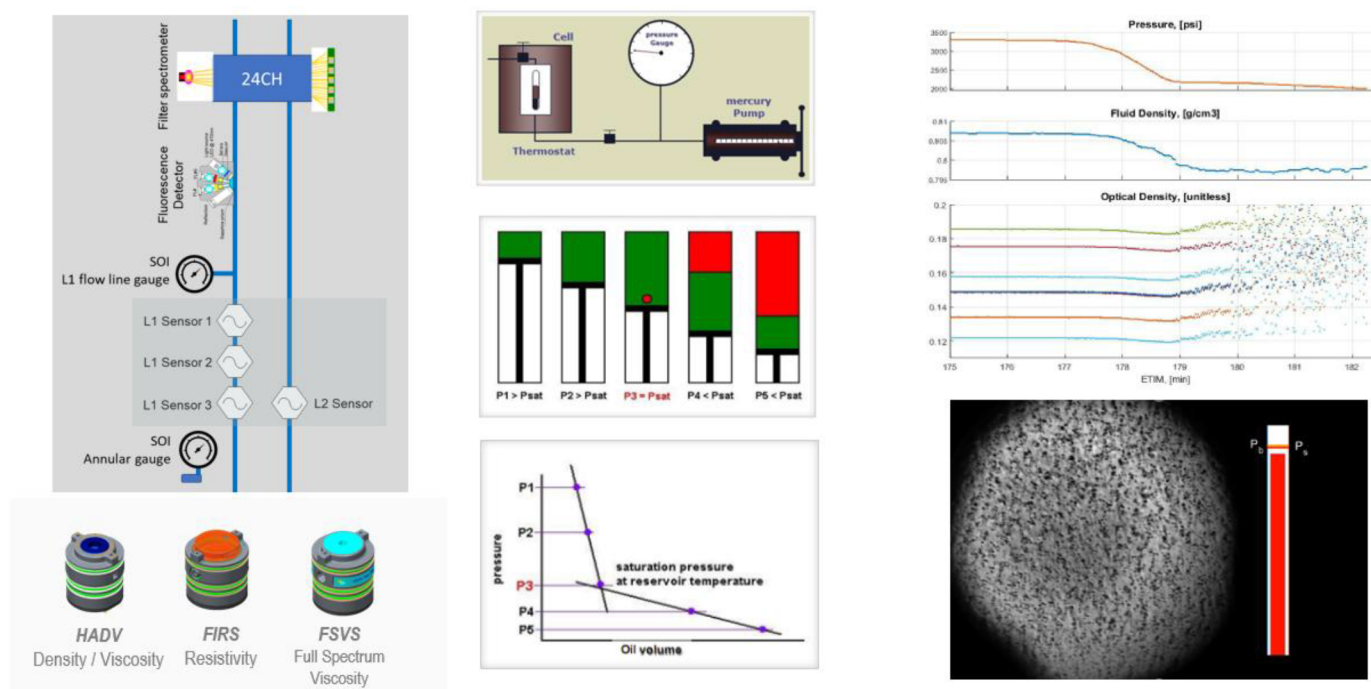


Figure 6—Fluid In-situ Scanner and Formation Testing Flowline Decompression Experiment to obtain Saturation pressure

Job Planning

A total of fourteen stations were strategically planned for this operation, comprising nine dedicated to fluid identification and five allocated for fluid sampling. These stations were distributed across one primary reservoir target and two secondary zones to ensure comprehensive dynamic reservoir evaluation. Traditionally, such operations are conducted using drill pipe conveyance, primarily due to the elevated risk of tool sticking in deviated wells. This risk is exacerbated by the extended stationary time required at each testing depth and the potential for significant hydrocarbon influx into the wellbore during clean-up. To mitigate these challenges and optimize operational efficiency, a detailed clean-up simulation was performed for each target zone. These simulations were essential for estimating the expected station time and calculating the total volume of hydrocarbons that would need to be pumped into the wellbore at each location. This proactive planning approach ensured safe and efficient execution of the formation testing program. (See Fig. 7)

A significant reduction in clean-up time was anticipated and successfully achieved through the integration of a focused radial inlet combined with a high-rate pump. This advanced configuration enabled the minimization of stationary time at each testing depth, bringing it down to as little as two hours per station. In comparison, legacy formation testing tools such as the Saturn inlet and Dual Packer systems require substantially longer clean-up durations—approximately 3.4 times and 4.8 times more pumping time, respectively—to reach the same level of fluid clarity necessary for accurate formation fluid identification.

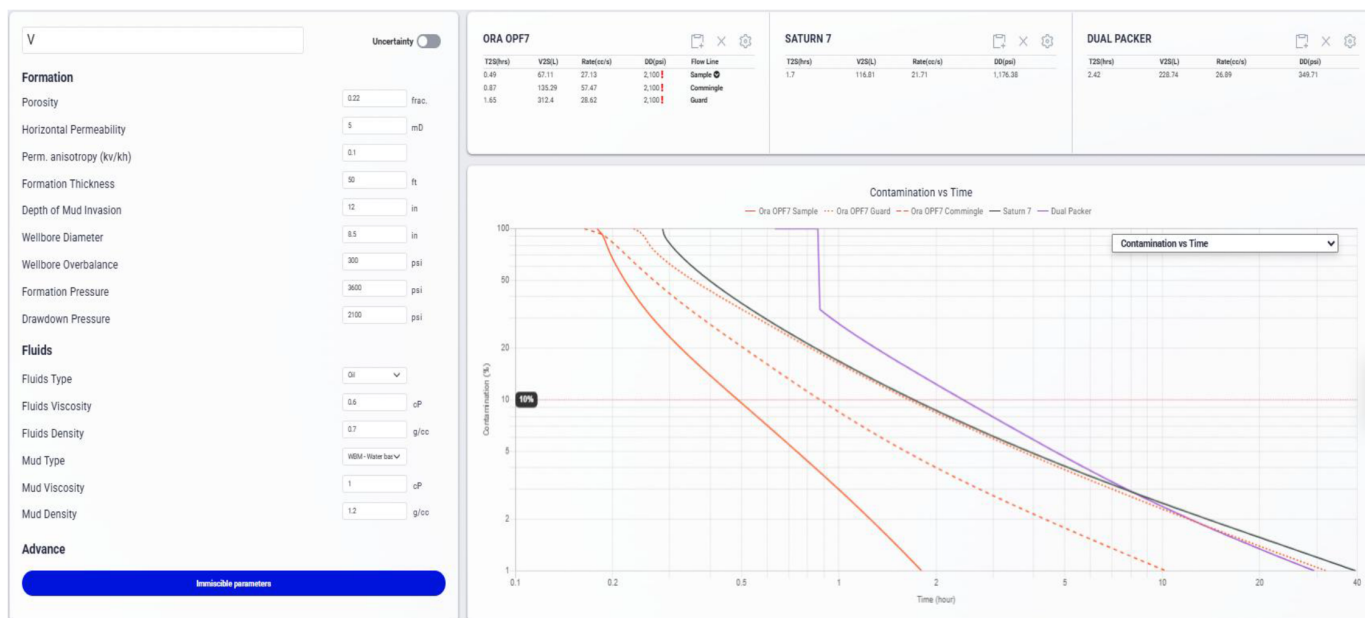


Figure 7—ORA Focus Radial Probe clean up simulation for main target zone estimating less than 0.5 hours clean up time with 67 liters volume per station.

This breakthrough in operational efficiency made it feasible to execute the entire formation testing program using wireline conveyance, eliminating the need for drill pipe deployment, which is traditionally preferred in deviated wells due to sticking risks. To further mitigate these risks, a high-tension wireline setup was employed, featuring a specialized ultra-strength cable and a heavy-duty logging unit. Tension simulations predicted a normal operating tension of 7,100 lbf, with an overpull capacity reaching up to 10,900 lbf, ensuring robust safety margins during tool conveyance (See Fig. 8).

Additionally, special rollers were incorporated into the toolstring design to reduce friction and facilitate smooth movement through the deviated wellbore. Wireline jars were also included as a contingency measure, capable of delivering up to 12,300 lbf of force if activation was required, further enhancing the reliability and safety of the operation.

One of the critical concerns during extended wireline formation testing operations is the volume of hydrocarbons that must be pumped into the wellbore during clean-up. This becomes particularly significant when the wireline tool is deployed in-hole, as the driller has limited control over well conditions. Specifically, if the hydrostatic pressure in the wellbore drops below the formation pressure, there is a risk of uncontrolled fluid influx or gas breakout at the surface, which can pose serious safety and operational challenges.

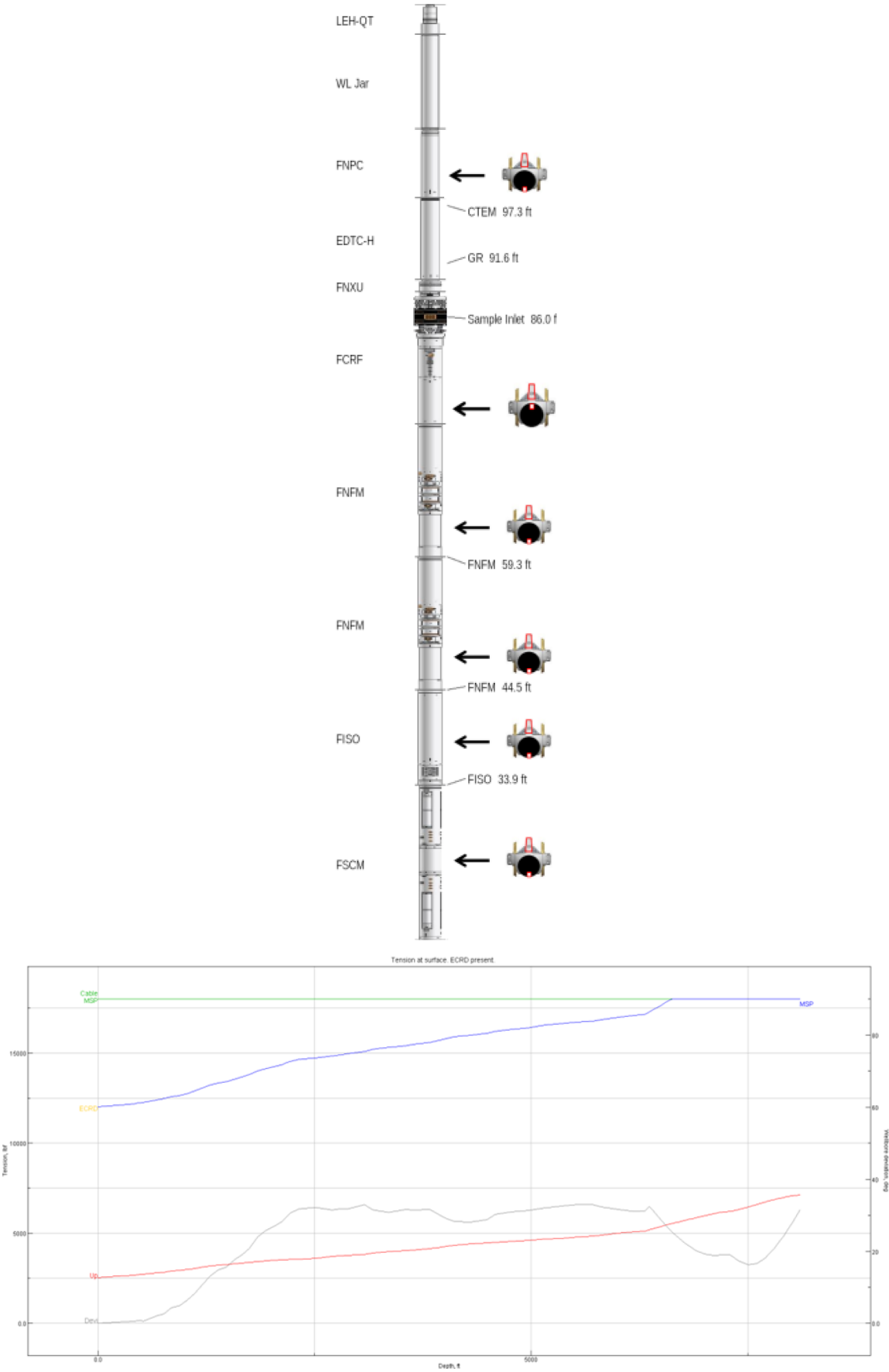


Figure 8—Tension simulation for ORA Toolstring with 18000 lbf. pull capacity Ultra High Strength Cable

To proactively address this risk, the ORA Intelligent Planning workflow incorporated a comprehensive wellbore dynamic simulation. This simulation was designed to model the maximum expected hydrocarbon influx during clean-up operations and assess its impact on wellbore pressure and surface conditions. In this particular case, the estimated volume of hydrocarbon to be pumped into the wellbore was approximately 566 liters (equivalent to 3.6 barrels). To ensure robustness, simulations were conducted not only for this expected volume but also for double that amount, providing a conservative safety margin.

The results of the simulation confirmed that the operation could proceed safely under both scenarios. The maximum pressure drop observed in the mud column was limited to 20 psi, which remained well within acceptable limits. Importantly, the simulation predicted no free gas at the surface, indicating that the wellbore would remain stable throughout the clean-up process. These findings validated the feasibility of the wireline-based approach and reinforced the importance of predictive modeling in managing operational risks in complex reservoir environments. (See Fig. 9)

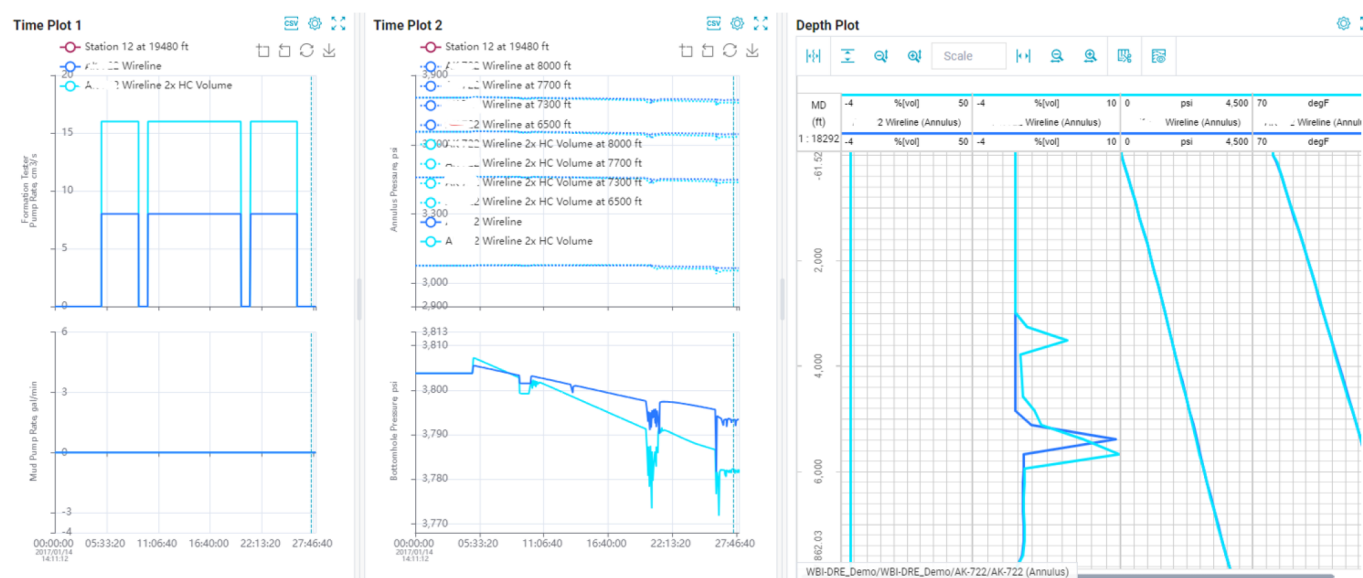


Figure 9—Wellbore Dynamic Simulation performed to estimate mud column pressure and free gas at surface

As part of the intelligent job planning process, the wireline formation testing operation was meticulously designed to optimize tool configuration and sampling efficiency. The toolstring was equipped with six sample bottles to accommodate the planned acquisition, with provisions made for a contingent trip in-hole should additional samples be required based on real-time fluid identification results. This level of flexibility ensured that the operation could adapt dynamically to subsurface conditions without compromising data quality or operational safety.

To further enhance efficiency and reduce operational risk, the stationary time at each testing station was deliberately limited to a maximum of two hours. In scenarios where extended clean-up was necessary to achieve representative fluid samples, the probe was programmed to retract and reset, allowing the clean-up process to continue without exceeding safe exposure limits. This technique is particularly effective when using radial and focused radial probes, as it has no dead volume between the packer and the formation, thereby clean up subsequent setting start where it left previous setting.

The use of focused radial inlet technology in conjunction with real-time lab accuracy metrology fluid identification enabled precise control over sampling parameters, ensuring that each station delivered high-quality data while maintaining operational integrity in a such challenging environment.

Job Execution

The wireline formation testing operation was executed with exceptional efficiency, successfully meeting all planned objectives within a single deployment. This achievement is particularly noteworthy given that actual reservoir permeabilities encountered during the operation were lower than initially anticipated, which typically poses challenges for fluid mobility and clean-up efficiency (see Fig. 10).

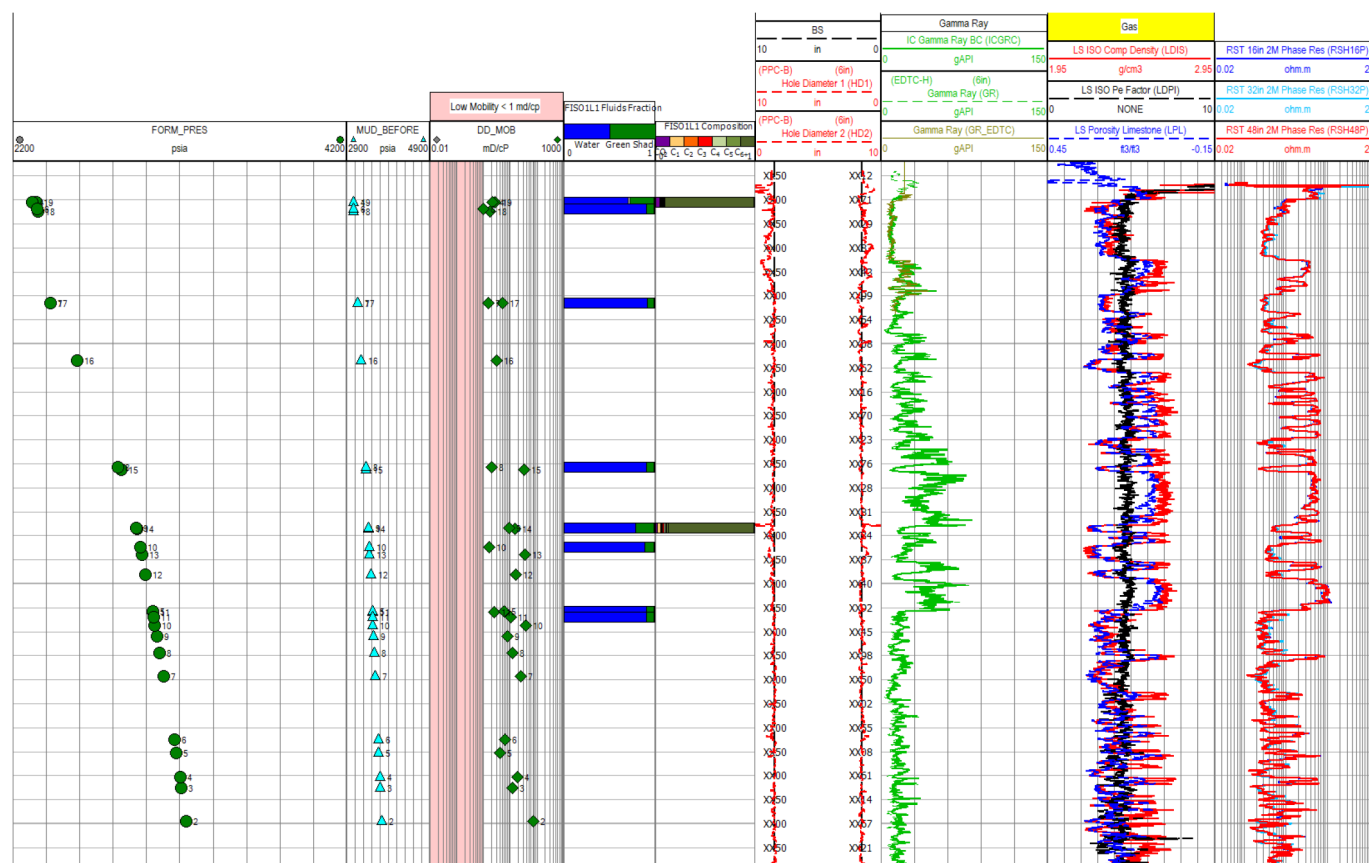


Figure 10—Depthview includes all pressure points with fluid identification and sampling stations

Despite these subsurface complexities, eight testing stations were completed in less than 24 hours, with each station requiring no more than two hours of stationary time. This level of performance significantly exceeds the operational capabilities of conventional formation testing tools, which often require extended durations for clean-up and fluid identification, especially in low-permeability carbonate formations.

The ability to complete the entire job in a single wireline run not only streamlined the workflow but also resulted in substantial operational savings. Specifically, it eliminated the need for drill pipe conveyance, which would have added approximately five additional days of rig time. This reduction in rig time translates into lower operational costs, reduced fuel consumption, and minimized environmental impact, reinforcing the value of deploying advanced wireline technologies like the ORA platform in offshore reservoir evaluation programs.

The deployment of the ORA Intelligent Wireline Formation Testing Platform yielded outstanding results, demonstrating its capability to deliver high-quality data acquisition in a single, streamlined operation. A total of six downhole fluid samples were successfully collected—four from two distinct oil-bearing zones and two from water-bearing intervals—providing a representative cross-section of the reservoir's fluid distribution. In addition to these samples, four supplementary fluid identification stations were executed, enabling real-time confirmation of formation fluid properties through advanced in-situ analysis. Moreover, radial flow

behavior was observed during the final pressure build-up at most stations, a key indicator of representative permeability and effective reservoir connectivity. This observation reinforces the reliability of the acquired pressure and fluid data, supporting robust reservoir characterization and dynamic modeling. (See Fig. 11)

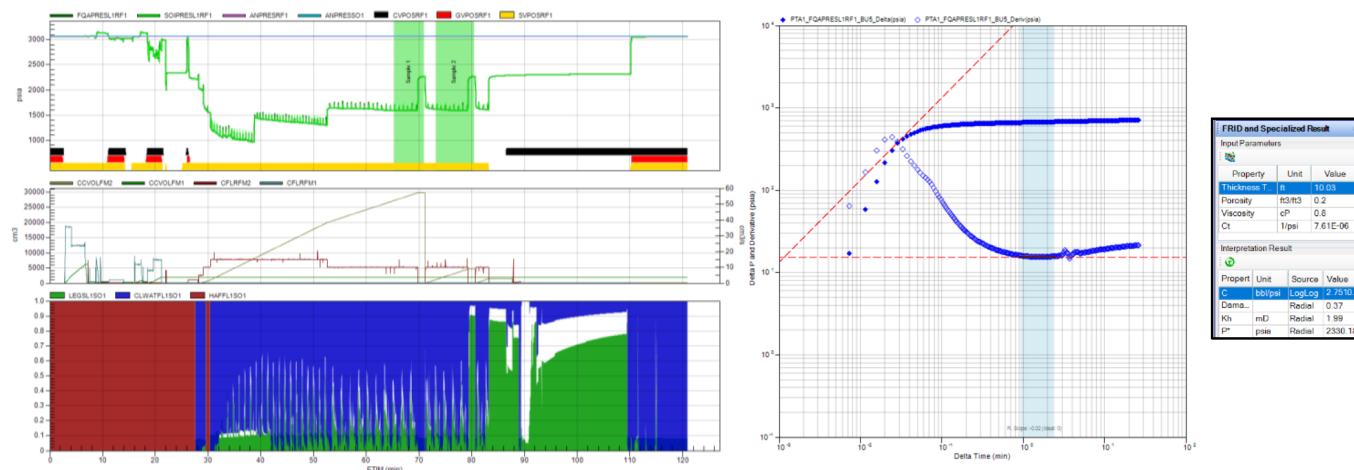


Figure 11—Pressure and fluid identification plots with final build up derivative

The embedded fluorescence detector within the ORA platform demonstrated exceptional sensitivity to hydrocarbons, proving its capability to detect even minute traces of oil in fluid flows dominated by water. This feature was particularly valuable in zones with high water saturation, where conventional tools often struggle to differentiate between formation water and residual hydrocarbons. The detector's ability to respond to subtle fluorescence signals enabled accurate fluid typing. Complementing this, the in-situ fluid analyzer provided a comprehensive suite of real-time measurements, including hydrocarbon composition, gas-oil ratio (GOR). These measurements were successfully acquired even in zones exhibiting water cuts as high as 80–90%, enjoying fluid segregation in the pump. Furthermore, saturation pressure was measured directly in the downhole environment using FISO sensors, but only in stations where sufficient hydrocarbon concentration was present in front of the sensor array. This capability allowed for accurate determination of bubble point pressure under reservoir conditions, contributing to a more complete understanding of phase behavior and fluid properties. (See Fig. 12)

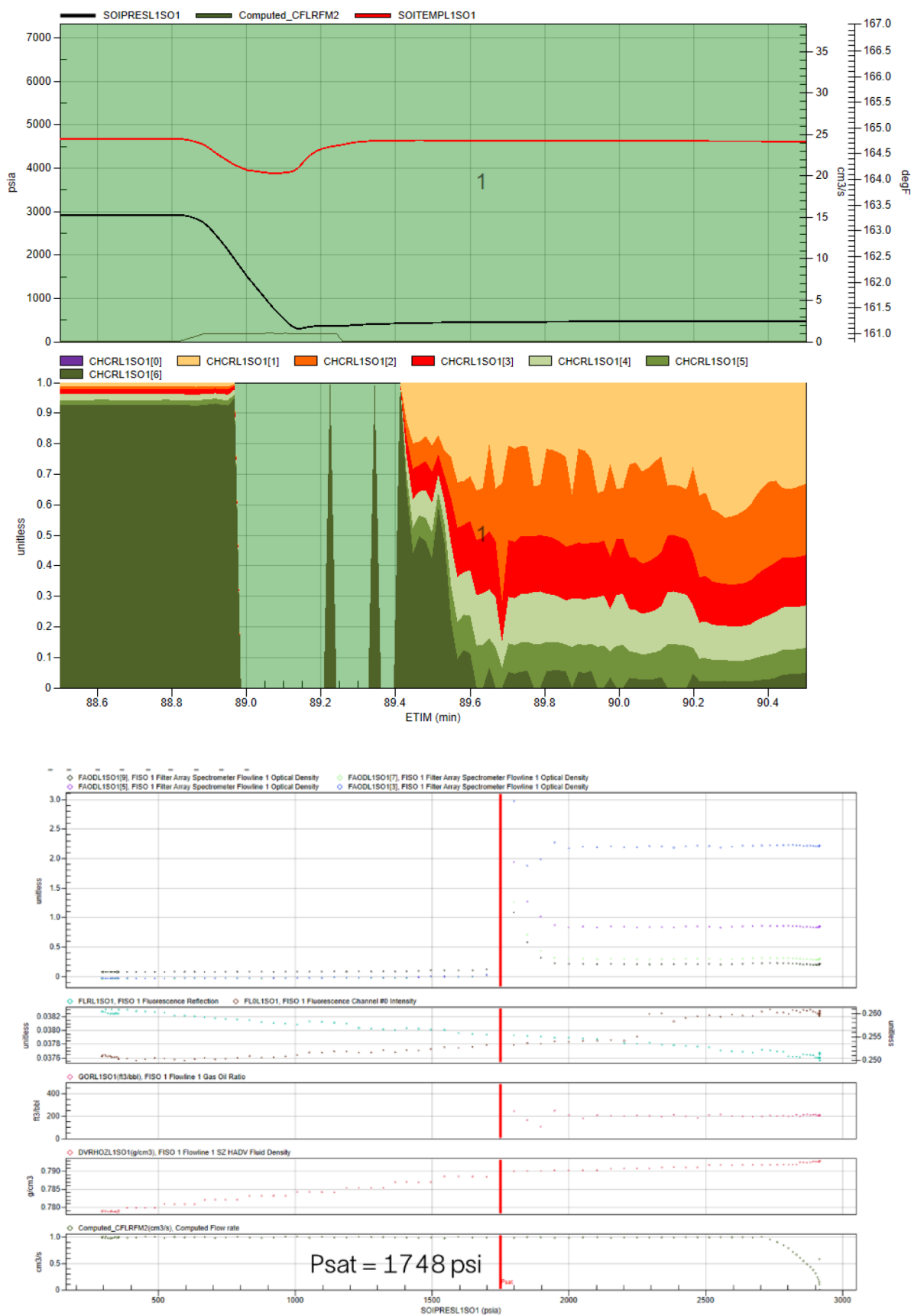


Figure 12—Saturation pressure measurement in downhole condition real time

Conclusion

This case study clearly illustrates the effectiveness of the ORA Intelligent Wireline Formation Testing Platform in addressing the complex challenges associated with evaluating low resistivity pay zones in carbonate formations, particularly within a deviated wellbore environment. The deployment of this advanced technology enabled high-resolution data acquisition through a single wireline run, despite the operational constraints typically encountered in such geological settings.

Central to this success was the platform's innovative dual-flowline architecture, which facilitated efficient clean-up and focused sampling. This was complemented by laboratory-grade metrology integrated into the in-situ fluid analyzer, allowing for precise real-time characterization of formation fluids. The system's advanced automation features, combined with a high-tension conveyance setup, ensured safe and reliable tool deployment, even in high-angle wells with elevated sticking risks.

The data acquired during the operation provided critical insights into the reservoir's fluid distribution, phase behavior, and pressure regimes. These findings enabled informed decision-making for field development planning, including reservoir modeling, production strategy, and well placement optimization. The ability to distinguish between oil-water contacts and identify transition zones with high confidence was particularly valuable for assessing reservoir connectivity and potential productivity.

Beyond technical performance, the operation demonstrated significant operational and environmental benefits. By completing the entire formation testing program in a single wireline run, the team avoided the need for drill pipe conveyance, resulting in a reduction of approximately five days of rig time. This translated into lower fuel consumption, reduced emissions, and minimized environmental footprint—aligning with ADNOC's sustainability objectives.

The lessons learned from this deployment serve as a benchmark for future applications in similar offshore carbonate reservoirs. They highlight the importance of integrating intelligent planning, advanced hardware, and real-time analytics to overcome subsurface uncertainties and deliver efficient, safe, and environmentally responsible reservoir evaluation practices.

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